Preaching Against Power Theft*

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Abstract

Can moral appeals improve compliance where electricity theft and non-payment undermine utility finances? We study two linked randomised field experiments in high-loss electricity networks in Pakistan. Across 42 feeders, we test repeated Friday sermons invoking moral obligations against theft and non-payment. Within sermon and control areas, we independently randomise bill subsidies among 1,219 households. Sermon assignment produces no detectable improvement in recorded feeder losses or collections, although estimates are imprecise and measured exposure is limited. By contrast, subsidies reduce households' cash payments by approximately PKR 500 per month without detectably increasing electricity consumption. In sermon areas, they also reduce outstanding arrears by PKR 2,808, with larger effects among households entering treatment in debt. The findings contrast attempts to change the perceived moral costs of non-compliance with direct financial relief that improves household balance sheets without increasing energy demand.

JEL codes: C93, D91, L94, Q41, Z12.

Keywords: electricity theft; non-payment; moral suasion; religion; subsidies; arrears; field experiment; Pakistan.

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1 Introduction

Electricity theft and non-payment weaken utilities in many developing countries, but policy responses face a basic question: are these behaviours best understood as failures of norms, failures of enforcement, or failures of affordability? These mechanisms imply different interventions. A moral appeal can attempt to raise the perceived social or ethical cost of non-compliance. A subsidy instead lowers the immediate private cost of paying for electricity. The distinction matters because both interventions can be deployed at scale, but through very different institutions and with very different fiscal consequences.

We study these mechanisms in Kasur district, Pakistan, in partnership with the Lahore Electric Supply Company (LESCO). The setting is one in which commercial losses are large and electricity service is politically salient. In the study sample, rural high-loss feeders had pre-treatment transmission and distribution losses of roughly 32 percent. The institutional environment also makes mosque-based communication unusually relevant: Friday prayer creates a repeated, locally embedded communication channel, and anti-theft messages framed in religious and social terms have previously been used by Pakistani electricity utilities.

The study combines two linked randomised experiments. In the first, 42 high-loss rural feeders were assigned equally to a mosque-sermon intervention or control. From November 2024 through April 2025, participating Friday-prayer mosques on treated feeders delivered repeated sermons discouraging electricity theft and non-payment. The feeder is both the unit of randomisation and the main unit of analysis. In the second experiment, 1,219 households from 50 selected locations were randomised within sermon and no-sermon strata to receive a bill subsidy or no subsidy. The subsidy was credited through the formal LESCO billing system and was designed to lower the amount households had to pay out of pocket.

The first result is that moral messaging alone does not measurably improve feeder-level commercial performance. Across the post-treatment period, the estimated effect on the recorded loss rate is -0.7 percentage points (SE 3.2), with a randomisation-inference p-value of 0.824. We likewise find no detectable effect on electricity received, billed units, assessments, or payments. Dynamic specifications do not reveal a sustained treatment response during or after sermon delivery. These estimates are imprecise nulls: the intervention used an institution with broad local reach, but shifting the moral framing of theft and non-payment was not sufficient to change aggregate feeder outcomes at the precision available in a 42-cluster experiment.

The second result is that the household subsidy changes debt and payment obligations without materially changing electricity consumption. In no-sermon locations, the subsidy reduces households' own cash payment by PKR 497 per month (SE 158) and has no detectable effect on consumption. In sermon locations, the subsidy is associated with a PKR 4,053 lower amount payable (SE 1,354), cash payment by PKR 511 (SE 166), and arrears by PKR 2,808 (SE 1,148), again with essentially no change in electricity use. The decline in cash payment is not evidence that the utility receives less resources dollar-for-dollar: the intervention itself credits part of the bill. Instead, it shows that the subsidy replaces part of the household's out-of-pocket payment rather than inducing higher

electricity demand.

A third result sharpens the debt interpretation. Within sermon locations, households that entered treatment with positive arrears experience especially large reductions in outstanding debt: the subsidy lowers the monthly arrears balance by PKR 7,256 (SE 2,777), compared with essentially zero among households without baseline arrears. The high-versus-low differential is -PKR 7,288 (SE 2,828). The same sharp heterogeneity is not present in no-sermon locations. Because subsidy effects are estimated separately by sermon stratum, comparisons across strata are descriptive rather than tests of complementarity. The baseline-arrears interaction within each stratum is identified by random subsidy assignment.

The paper contributes to three literatures. First, it adds experimental evidence on moral and religious persuasion in economic behaviour, complementing work showing that moral appeals can change repayment or other choices in some settings (Bursztyn et al., 2019; Bryan et al., 2021). Second, it speaks to the political economy of electricity provision in South Asia, where norms around payment and service quality can sustain low-recovery equilibria (Burgess et al., 2020). Third, it contributes to work on energy subsidies by showing a setting in which financial relief changes balance-sheet outcomes without a detectable increase in electricity quantity. The combination is useful precisely because the two experiments act on different margins: moral suasion attempts to change the willingness to comply, while the subsidy changes the immediate financial constraint.

The rest of the paper proceeds as follows. Section 2 describes the setting and treatments. Section 3 presents the experimental design and data. Section 4 explains the empirical strategy. Section 5 reports the feeder-level sermon results and Section 6 the household subsidy results. Sections 7–9 develop interpretation, additional evidence and discussion; Section 10 concludes.

2 Setting and Interventions

2.1 Electricity losses and collections in Pakistan

Pakistan's distribution companies face losses on two distinct margins. First, electricity delivered to a network can exceed electricity billed to consumers because of technical losses, illegal connections, meter bypass, tampering, and administrative measurement error. Second, even when electricity is billed, utilities may fail to collect the full amount due. We refer to the first as recorded distribution losses and the second as a collection shortfall.

In fiscal year 2023–24, Pakistan's distribution companies reported weighted-average transmission and distribution losses of about 18 percent, above the allowed average, while recovery of billed amounts remained below full collection (NEPRA, 2024a,b). These shortfalls contribute to utility financial stress and the electricity sector's broader circular-debt problem. The policy challenge is therefore not simply to reduce electricity consumption, but to bring more consumption into the formal billing and payment system.

The study is conducted in Kasur district, within LESCO's service territory. Kasur is predominantly rural and its electricity performance historically lagged LESCO as a whole. We focus

on rural feeders that were classified as high loss in at least one month of 2022 or 2023. After administrative, geospatial, and overlap screens, the experimental sample contains 42 feeders with a mean pre-treatment recorded loss rate of approximately 32 percent.

2.2 Mosque-based moral messaging

Friday prayer provides a repeated, locally embedded communication institution. In Pakistan, mosques can facilitate information diffusion and community coordination, while Friday sermons create a predictable setting in which an imam can frame behaviour in moral terms (Cheema et al., 2014; Butt, 2016). Electricity utilities have themselves appealed to religious authority in anti-theft campaigns (Business Recorder, 2009). The intervention therefore tests whether a moral framing can shift behaviour through an institution already present in the relevant communities.

Twenty-one feeders were assigned to the sermon arm and 21 to control. Within treated feeder catchment areas, implementation teams identified up to two of the largest Friday-prayer mosques in each treated location. From November 2024 through April 2025, anti-theft and payment messages were delivered repeatedly through sermons. The content emphasised that electricity theft and deliberate non-payment harm other users and violate an obligation to pay for a service. No analogous sermon programme was implemented on control feeders.

2.3 Household bill subsidies

The second experiment was nested within 50 selected locations: 25 in sermon areas and 25 in no-sermon areas. Survey teams enrolled 617 households in sermon locations and 602 in no-sermon locations. Within each stratum, households were randomised to subsidy or no subsidy. Eligibility required formal billing information and relatively modest recent electricity consumption.

The subsidy was integrated into LESCO's official billing system, so treated households saw a lower amount payable on the electricity bill itself rather than receiving a later cash transfer. From November 2024 through May 2025, the programme subsidised 30 percent of the current bill, capped at PKR 2,000. From June through August 2025, it subsidised 50 percent, capped at PKR 4,000. The intervention ended in August 2025. This structure creates a direct reduction in households' immediate payment obligation while preserving the posted electricity tariff and formal billing relationship.

3 Experimental Design and Data

3.1 Design

Figure 1 summarises the two-stage design. The sermon experiment is randomised at the feeder level. The household subsidy experiment is randomised at the household level within sermon and no-sermon strata. The latter therefore identifies subsidy effects separately in two experimentally created information environments, but the sampled household locations themselves were selected

purposively from the broader feeder catchments. Household estimates should thus be interpreted as internally valid for the sampled households rather than as design-based estimates for all households on the 42 feeders.

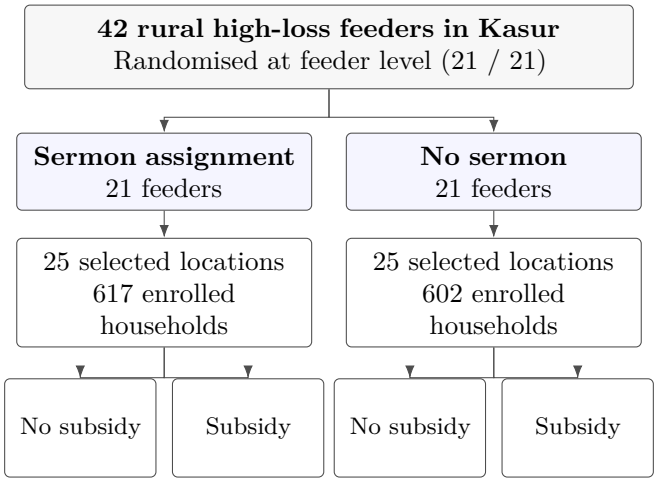


Figure 1: Experimental design: feeder-level sermons and nested household subsidy randomisation

Notes: Feeder assignment was randomised 21/21. Twenty-five locations per feeder arm were selected non-randomly for the household study; 617 and 602 households were enrolled, respectively. Subsidy assignment was randomised within each of these two samples. The terminal boxes represent the four experimental cells, not four separately randomised feeder groups.

3.2 Feeder administrative data

The feeder panel uses monthly LESCO administrative records from January 2021 through September 2025. For most outcomes the panel contains 2,394 feeder-month observations before sample restrictions in particular specifications. Core outcomes are electricity received, electricity billed, units lost, the recorded loss rate, assessed bill amounts, and payments. The primary feeder specification uses the post period beginning in November 2024, the start of the sermon intervention.

The recorded loss rate is a network accounting outcome, not a direct measure of theft. It can incorporate technical losses, illegal consumption, meter and mapping errors, and timing mismatches in administrative records. For that reason, we describe treatment effects as effects on *recorded feeder losses* and do not equate a reduction in the loss rate mechanically with the same reduction in theft.

3.3 Household administrative and survey data

The household panel contains 1,219 households observed monthly from January 2023 through November 2025, yielding 41,787 household-month observations in the raw panel. Administrative outcomes include electricity use, amount payable within the due date, amount paid by the consumer, arrears, and subsidy received. We define household collection loss as one minus the ratio of payment to the amount payable within the due date.

Baseline survey measures include solar adoption, observed hooked connections, religiosity, Friday-prayer participation, income, expenditures, and housing characteristics. Endline survey

data include self-reported sermon exposure, solar adoption, and observed hooked connections. The survey measures allow us to check balance, describe implementation, and study pre-specified or economically motivated heterogeneity.

4 Empirical Strategy

4.1 Feeder-level sermon effects

For feeder f in month t, we estimate

$$Y_{ft} = \alpha_f + \lambda_t + \beta \left(\text{Sermon}_f \times \text{Post}_t\right) + \varepsilon_{ft}, \tag{1}$$

where α_f and λ_t are feeder and month fixed effects and Post_t begins in November 2024. Standard errors are clustered at the feeder, the unit of randomisation. Because there are only 42 feeders, we also report randomisation-inference p-values obtained by repeatedly reassigning 21 of the 42 feeders to placebo treatment.

We complement the average post-treatment effect with period-specific estimates for November 2024–January 2025, February–April 2025, May–July 2025, and August–September 2025. The first two bins cover the active sermon period and the latter two provide evidence on persistence after sermon delivery ends.

4.2 Household subsidy effects

Because subsidy assignment was randomised separately within sermon and no-sermon strata, we estimate household fixed-effects models separately by stratum:

$$Y_{it} = \alpha_i + \lambda_t + \tau \left(\text{Subsidy}_i \times \text{Post}_t\right) + \varepsilon_{it}. \tag{2}$$

Standard errors are clustered at the household. We also estimate time-varying effects and post-period cross-sectional specifications. For heterogeneity by baseline arrears, we interact treatment with an indicator for positive arrears in October 2024 and allow the high- and low-arrears groups to follow separate month effects.

A key interpretive point is that the current analysis estimates subsidy effects separately within the sermon and no-sermon strata. We therefore do not treat differences between the two stratum-specific coefficients as statistically established complementarity unless a direct cross-stratum interaction test is reported. By contrast, heterogeneity within a stratum — for example, subsidy assignment interacted with pre-treatment arrears — retains the experimental interpretation because the moderator is measured before treatment.

5 Effects of Mosque-Based Moral Messaging

5.1 Average feeder outcomes

Table 1 reports the main feeder estimates. Sermon assignment has no detectable effect on any of the core network outcomes. The recorded loss-rate estimate is -0.0074 (SE 0.0319), less than one percentage point relative to a pre-treatment mean of roughly 32 percent. Randomisation inference produces a p-value of 0.824. The point estimates for received and billed units, assessments, and payments are likewise small relative to their standard errors and have large randomisation-inference p-values.

Table 1: Effect of mosque-sermon assignment on feeder outcomes

	Received units (MWh)	Billed units (MWh)	Recorded loss rate	Assessment (million PKR)	Payment (million PKR)
Sermon × Post	-4.81	27.3	-0.0074	0.105	-0.205
	(59.5)	(40.5)	(0.0319)	(1.09)	(1.01)
RI p-value	0.933	0.533	0.824	0.931	0.847
Observations	2,388	2,388	2,359	2,359	2,388
Feeder FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes

Notes: Two-way fixed-effects regressions comparing 21 sermon-assigned feeders with 21 control feeders. Standard errors clustered at feeder level in parentheses. RI p-values preserve the 21/21 treatment split over 4,999 placebo re-randomisations. The recorded loss outcome is an administrative network-accounting measure and should not be interpreted as a direct measure of theft.

The null is not confined to a single post-treatment average. Period-specific estimates move around zero across the active treatment and persistence windows. For the recorded loss rate, the point estimates are +2.2 percentage points in November–January, -5.3 points in February–April, -6.6 points in May–July, and -2.1 points in August–September, each estimated imprecisely. The pattern offers no evidence of a stable reduction in recorded losses that begins with sermon delivery and persists thereafter.

5.2 Implementation and interpretation

The null feeder result can arise either because exposure was insufficient or because exposure did not change behaviour. The household endline offers one implementation check: self-reported exposure to a relevant sermon is 45.6 percent in sermon locations and 40.6 percent in no-sermon locations, a 5.0 percentage-point difference (SE 2.9). That first stage is statistically weak and substantively modest. Self-reports may of course be noisy, and the feeder intervention reached a broader population than the nested survey sample. Still, the limited measured exposure difference cautions against interpreting the feeder null as a clean rejection of all forms of moral persuasion. The evidence does not detect a feeder-wide response to the campaign *as implemented*; the confidence interval still permits economically meaningful effects.

6 Effects of Household Bill Subsidies

6.1 Average effects

Table 2 reports the household fixed-effects estimates separately by sermon environment. Three patterns stand out. First, electricity consumption barely responds: the point estimate is about 2–3 kWh per month in both strata, with standard errors of roughly 3 kWh. Second, households' cash payment falls by approximately PKR 500 per month in both strata. Third, in sermon locations, the subsidy substantially lowers the amount payable and the stock of arrears. The arrears effect is -PKR 2,808 per month (SE 1,148), close to the pre-treatment mean arrears of PKR 2,704 in that stratum.

Table 2: Household subsidy effects, by sermon environment

	Units (kWh)	Payable (PKR)	Payment (PKR)	Collection loss rate	Arrears (PKR)
Panel A: No-sermon locations					
Subsidy × Post	2.28	-2,381	-497***	-0.0102	-1,378
	(3.12)	(1,807)	(158)	(0.0186)	(1,672)
Pre-period mean	111	6,068	2,963	0.259	2,802
Observations	20,616	20,616	20,616	20,607	20,616
Panel B: Sermon locations					
Subsidy × Post	2.73	-4,053***	-511***	-0.0191	-2,808**
	(3.20)	(1,354)	(166)	(0.0163)	(1,148)
Pre-period mean	111	5,911	2,889	0.244	2,704
Observations	21,171	21,171	21,171	21,168	21,171

Notes: Household and month fixed-effects regressions, estimated separately within the no-sermon and sermon strata. Standard errors clustered at household level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table does not report a formal test of equality of the subsidy coefficients across sermon strata.

The combination of lower arrears and unchanged consumption is important. A conventional concern with subsidising electricity is that it may lower the effective marginal price and increase quantity demanded. We do not see that response here. Instead, the subsidy primarily changes the financial position of households: it lowers the amount they must finance and, in the sermon stratum, reduces accumulated debt. This is consistent with an intervention operating through liquidity or balance-sheet relief rather than through a large quantity response.

The fall in household cash payment should be interpreted together with the subsidy credit. Treated households receive part of their bill through the programme, so a reduction in the household's own payment does not map one-for-one into a decline in utility receipts. The administrative treatment check confirms substantial subsidy receipt in both strata, with larger average credits in the later, more generous phase.

6.2 Dynamics

Figure 2 shows that the lack of a detectable consumption response is not an artefact of averaging. Event-study estimates fluctuate around zero through the treatment period in both strata, with

pre-treatment joint tests that do not reject parallel trends. The post-subsidy coefficients are also small relative to their uncertainty.

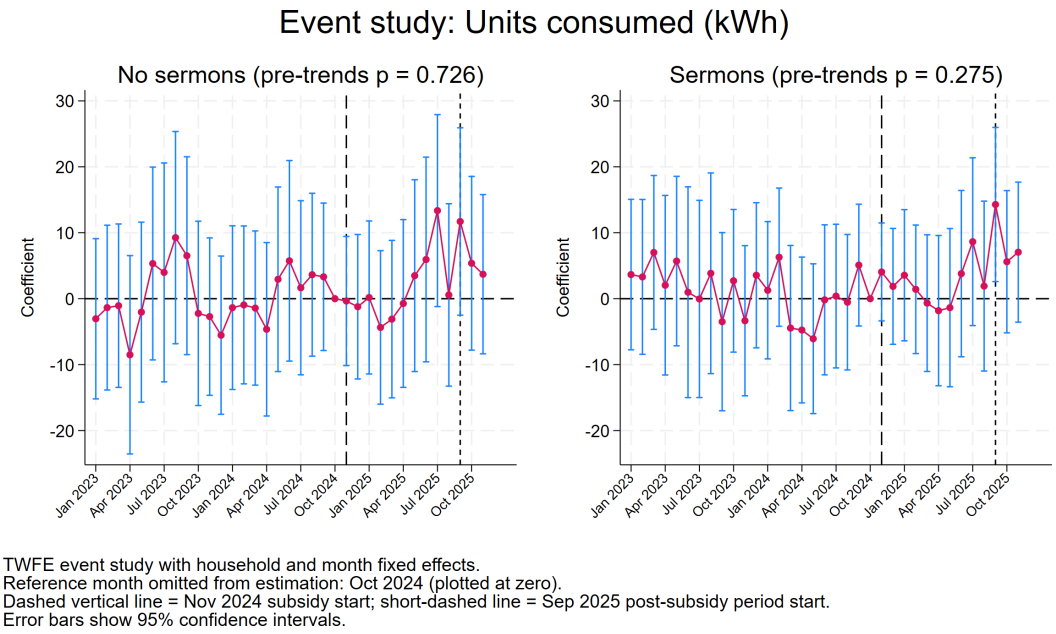


Figure 2: Dynamic effects of the household subsidy on electricity consumption

Notes: Household event-study estimates shown separately for no-sermon and sermon strata. The omitted month is October 2024. Vertical lines mark the start of the subsidy and the post-subsidy period. Error bars show 95 percent confidence intervals.

Arrears display a different pattern. Figure 3 shows negative post-treatment coefficients emerging in both strata, with larger and more persistent point estimates in sermon locations. The time-series evidence is consistent with the average treatment effects in Table 2: the intervention changes debt more clearly than it changes electricity use.

6.3 Who experiences the debt reduction?

Table 3 splits households by whether they had positive arrears immediately before treatment. In no-sermon locations, the subsidy effect on arrears is statistically indistinguishable from zero for both groups. In sermon locations, the effect is highly concentrated among households that entered treatment already in debt. Their average monthly arrears balance is PKR 7,256 lower (SE 2,777), and the difference relative to households without baseline arrears is -PKR 7,288 (SE 2,828). The same group also sees a large decline in the amount payable.

This heterogeneity is consistent with the subsidy relaxing a debt constraint rather than simply increasing contemporaneous electricity demand. It also shows why average arrears can move more than consumption: households can use the subsidy to prevent new debt or reduce outstanding balances while holding electricity use approximately fixed. Although the point estimates differ across sermon environments, these separate within-stratum estimates do not establish an interaction.

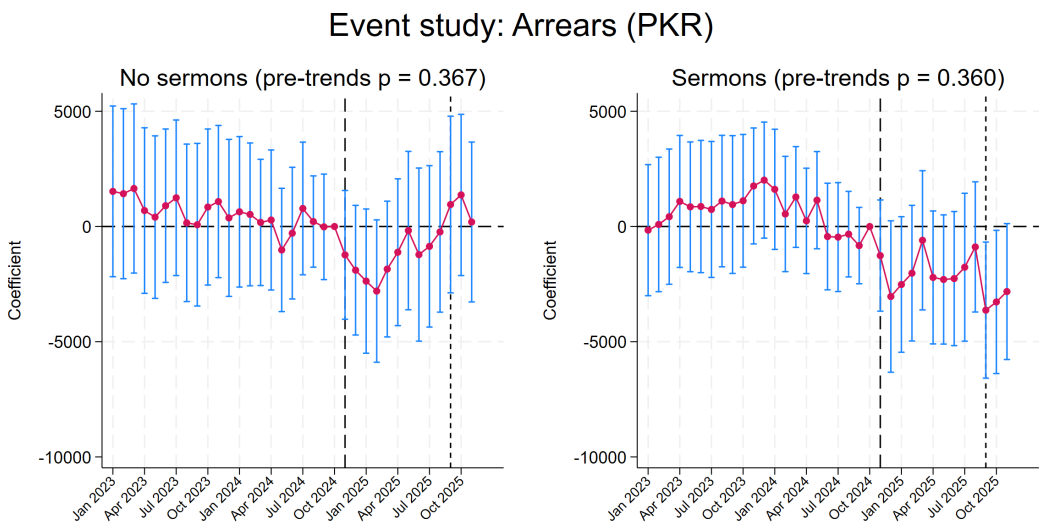


Figure 3: Dynamic effects of the household subsidy on arrears

Notes: Household event-study estimates shown separately for no-sermon and sermon strata. The omitted month is October 2024. Error bars show 95 percent confidence intervals.

Table 3: Subsidy effects by baseline arrears status

	Payable (PKR)	Payment (PKR)	Arrears (PKR)
Panel A: No-sermon locations			
No baseline arrears: subsidy effect	-3,175**	-529***	-2,184
	(1,583)	(193)	(1,427)
Positive baseline arrears: subsidy effect	-1,072	-438*	-97
	(3,578)	(262)	(3,348)
High − Low differential	2,104	90.6	2,088
	(3,913)	(325)	(3,639)
Panel B: Sermon locations			
No baseline arrears: subsidy effect	-706	-451**	31
	(822)	(187)	(537)
Positive baseline arrears: subsidy effect	-9,297***	-613**	-7,256***
	(3,173)	(308)	(2,777)
High − Low differential	-8,591***	-162	-7,288**
	(3,278)	(360)	(2,828)

Notes: Household fixed-effects regressions. "Positive baseline arrears" indicates positive arrears in October 2024. Models allow high- and low-arrears households to follow separate month effects. Standard errors clustered at household level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A direct cross-stratum test would be required.

7 Interpreting the Two Experiments

7.1 The extensive and intensive margins of compliance

A distribution utility can experience a fall in revenue for several reasons. First, customers may physically consume electricity without recording it on a meter, through hooked connections or meter bypass. Second, recorded consumption may be billed but remain unpaid. Third, even in the absence of theft, technical losses and administrative mismatches can create a gap between energy entering a feeder and energy attributed to consumers. These distinctions are central to the paper: a sermon intended to alter the perceived wrongness of theft may affect the first two margins, whereas the subsidy directly changes the amount due from an enrolled customer. Neither intervention should automatically be expected to move every accounting outcome in the same way.

Let R_{ft} be electricity received by feeder f, and B_{ft} billed energy. The recorded energy-loss share $L_{ft} = (R_{ft} - B_{ft})/R_{ft}$ depends on both its numerator and denominator. Thus an estimate close to zero for L_{ft} cannot distinguish a genuinely unchanged level of unauthorised consumption from offsetting changes in technical losses, billed energy, or records. Looking across the separately measured components is useful: our estimates for received units, billed units and the total unbilled gap are themselves imprecise, and hence do not point toward a clear alternative aggregate response to the sermon assignment.

Financial outcomes require still more care. The assessed amount is the utility's billed financial claim, payments are transfers credited against that claim, and arrears reflect accumulated unpaid obligations. At the household level the post-treatment subsidy is recorded as a separate bill credit. A household's choice to pay less cash may therefore be consistent with a reduction in its own obligation rather than an increase in delinquency. These outcomes do not allow us to identify households' counterfactual allocation of each rupee of relief between debt reduction, spending and saving.

7.2 What the sermon null does and does not establish

The estimated sermon effect on recorded aggregate losses is -0.74 percentage points, with a standard error of 3.19 percentage points. A conventional approximate 95 percent interval stretches from a reduction of about seven points to an increase of roughly five and a half points. With 42 clusters, the experiment does not rule out every potentially relevant effect. The measured endline exposure difference of approximately five percentage points also raises a separate implementation issue: an intention-to-treat effect averages over all feeder customers, not merely those who heard an intervention sermon. It therefore reflects the combined effects of message delivery, exposure, persuasion and any downstream changes in behaviour. The endline self-report should not be interpreted as an exact reach rate for every treatment mosque or as evidence that treated audiences uniformly accepted the message.

The assignment was at feeder rather than mosque or household level. That is appropriate for the utility outcomes: electricity metering, billing and revenue are observed for the entire feeder. But it can produce attenuation when only a limited subset of feeder customers receives the message. Leakage into control locations is also possible because mosque attendance and conversations need not respect administrative feeder boundaries. Screening overlapping feeders at baseline reduces this concern without proving the absence of such spillovers. What is experimentally identified is the aggregate effect of assignment to the specific campaign that was delivered.

7.3 What the household experiment identifies

The household lottery is independent within the two sermon environments. Its cleanest comparisons are therefore subsidy-assigned versus subsidy-control households *within* either environment. The estimated effect in a sermon area combines the subsidy response in that sampled setting with anything else specific to households there; it is not by itself the causal effect of adding a sermon to the subsidy. A stronger statement about complementarity would require a pooled interaction or equivalent formal comparison of the two lottery effects with uncertainty computed at the appropriate level for feeder-level sermon assignment. The estimates reported here do not establish that interaction.

Nor is the bill credit a uniform per-kWh price cut. It was assessed on the current bill, subject to a percentage and a cap, so the relationship between additional consumption and the marginal subsidy depends on each household's bill and whether the cap binds. The low quantity response is consequently evidence about *this schedule* rather than a general rejection of household price sensitivity. It is particularly notable alongside the strong reductions in payment obligations and selected debt measures.

7.4 Household exposure, selection and inference

Baseline and endline household surveys supplement the administrative panel, but the surveyed locations were purposively chosen after the feeder randomisation. The household treatment effects remain experimentally interpretable for the enrolled population because subsidy assignment took place after enrolment; they are not automatically representative of all customers on treated or control feeders. Households had to provide a bill and had to satisfy an electricity-consumption eligibility rule, selecting relatively modest-usage customers in a low-income rural setting. Interpretation of the heterogeneity results should stay local to this population.

Because the household subsidy was assigned to customers, its reported standard errors are clustered by customer. By contrast, direct comparisons between sermon environments would need to account for the feeder-level assignment and the small number of feeders. Some baseline covariates differ between household lottery arms by chance, as shown in the appended balance tables. The panel fixed effects address level differences in observed outcomes; the adjusted post-period specifications offer a separate check but do not replace the randomisation.

8 Additional Evidence on Dynamics and Heterogeneity

8.1 Feeder outcomes beyond the pooled treatment effect

The additional feeder specifications are checks on the same randomised intervention, not independent experiments. The randomisation-inference results for the principal feeder specifications use 4,999 placebo allocations and give little support for a detectable response across measured energy and financial outcomes. The period-specific estimates occasionally have sizeable point estimates; for example, the estimated loss-share difference is negative in the middle of 2025. Those movements have wide intervals, arise amid multiple outcomes and periods, and do not establish a lasting response. The event-study figures for assessments, payments, received units and billed units likewise show substantial uncertainty.

The feeder heterogeneity specifications split feeders by baseline recorded loss, connected load or consumer count, and one analysis compares treated feeders with high versus low mosque intensity. These are exploratory analyses on only 42 experimental units. In particular, mosque intensity is measured among the treated feeders and was not itself experimentally assigned. A smaller estimated coefficient in a high-intensity group should not be read as an experimental dose-response curve. Selected relevant estimates, including standard errors, are presented in the appendix.

8.2 Subsidy implementation and payment dynamics

The administrative delivery check is especially helpful in separating a weak outcome response from failure to implement a treatment. The programme credited treated households an average of about PKR 510 (no sermons) or PKR 459 (sermons) monthly during its first phase, with no corresponding subsidy among controls. Under the more generous phase, these amounts rose to about PKR 1,844 and PKR 1,580. This substantially more direct financial first stage contrasts with the modest survey-based separation in sermon exposure. These are descriptions of separately randomised treatments and implementation measures, not proof that one intervention would always outperform the other on a common welfare metric.

In the time-disaggregated household regressions, cash payments are consistently lower during subsidy receipt in the no-sermon arm, while the apparent reductions are larger during the later subsidy phase in both sermon environments. The end-of-programme periods are also informative. In sermon locations, the estimated arrears difference remains negative in September–November 2025 even though the measured subsidy credit falls to essentially zero. In no-sermon locations the analogous arrears estimate is close to zero. Both are contrasts within the respective lottery strata; the difference between these late-period coefficients is not itself accompanied by a statistical test. The fuller results and event studies appear in the appendix.

8.3 Financial exposure before treatment

Two alternative measures of financial vulnerability show a related pattern. First, positive arrears immediately before the programme identify households already carrying unpaid balances; this is the main heterogeneity comparison reported in Section 6. Second, we classify customers according to whether their current bill exceeded PKR 6,667 in at least ten percent of pre-treatment months. Within sermon locations, households meeting that persistent-high-bill definition have large negative point estimates for the amount payable, household cash payments and arrears. The estimated differential in arrears between high- and low-bill households is negative and imprecisely estimated relative to its size. The bills/arrears patterns are economically consistent with a subsidy relieving pressure where pre-existing financial obligations are larger, though the two indicators select related rather than identical groups.

Third, budget stress is defined by baseline reported expenses relative to income. In sermon locations the estimates indicate an arrears effect of roughly −4,162 PKR for the lower-stress subgroup versus a small positive point estimate for the higher-stress subgroup; the high-minus-low differential is positive. This should temper any single-factor liquidity story. Finally, baseline Friday-prayer attendance exposure, which might be expected to predict sermon reach, does not show a clear corresponding pattern in subsidy effects. It is particularly important not to present these many subgroup regressions as a sequence of independently confirmed mechanisms. They are exploratory; selected tables with coefficients and uncertainty appear in the appendix.

8.4 Robustness from alternative presentations

The post-period cross-sectional household specifications compare mean post-programme outcomes by lottery assignment, with and without baseline covariates and prior mean outcomes. Their role is to verify whether the main household conclusions depend on the fixed-effects representation of the monthly panel. In the adjusted sermon-location sample, the arrears effect is approximately PKR −2,713 (SE 1,156), close to the panel estimate of PKR −2,808 (SE 1,148). The household consumption effects remain small and statistically imprecise. The cash-payment coefficient stays negative in both strata. Sample sizes do change when baseline survey covariates are required, and the adjusted cross-sectional results must be read with that qualification. The appended tables provide the exact estimates and samples.

8.5 Limits to welfare and policy conclusions

Neither experiment measures all terms in social welfare. A sermon campaign has organisational and monitoring costs even when its aggregate utility outcomes do not move. The subsidy is a fiscal transfer whose social value depends on who bears its cost, which households benefit, and whether the credit leads to additional collection by the utility or merely replaces cash that households would otherwise pay. Without programme cost and utility accounting reconciliation, it would be unjustified to rank the two interventions by benefit–cost ratios. Likewise, lower household arrears is

not automatically an equal increase in utility cash collections. The appropriate policy takeaway is narrower: these data provide no measurable feeder-wide response to the campaign as delivered, while an official bill subsidy changes household financial outcomes without a detectable increase in recorded electricity use.

9 Discussion

The paired experiments deliver a useful contrast between changing beliefs or norms and changing household finances. The mosque campaign attempted to shift the perceived moral cost of non-compliance through an institution with regular access to local communities. The feeder results show that this intervention, as implemented, did not measurably alter aggregate losses or collections. The household experiment instead changed the budget constraint directly and reduced out-of-pocket payments without a detectable consumption response.

There are four limitations to the interpretation. First, recorded feeder losses are not a pure theft measure. They combine technical and commercial losses and may contain administrative noise. Second, the feeder experiment has only 42 clusters, so moderate effects cannot be ruled out even though the point estimates are small. Third, measured sermon exposure at endline differs only modestly across the household samples, suggesting imperfect reach or noisy recall. Fourth, the household sampling locations were selected purposively from within the feeder arms; the household causal estimates are therefore local to the sampled population.

The absence of a detectable consumption response is consistent across average and dynamic specifications and is substantively important. It implies that, in this sample and over this horizon, the subsidy did not generate a detectable quantity increase. Instead, it appears to shift who finances the bill and whether arrears accumulate. That distinction matters for subsidy design: financial support targeted at indebted households may improve household balance sheets without necessarily worsening physical electricity demand.

The results also caution against assuming that communication through a salient social institution automatically changes behaviour. The mosque is a plausible and scalable messenger, but the relevant behavioural margin may be constrained by affordability, enforcement, or limited message reach. A campaign that changes stated norms but not the material incentives surrounding electricity use and payment may be insufficient to move administrative outcomes.

10 Conclusion

We test two ways of addressing weak payment and high recorded electricity losses in rural Pakistan. A six-month mosque-sermon campaign designed to increase the moral cost of theft and non-payment produces no detectable improvement in feeder-level losses, billing, or payments. A household bill subsidy, by contrast, reduces out-of-pocket payments in both strata and recorded arrears in sermon locations, without a detectable increase in electricity consumption. Debt reductions are especially

large among households entering treatment with arrears in sermon locations.

The central distinction is between quantity and finance. The subsidy does not detectably increase electricity use; instead, it changes the household balance sheet. The moral campaign, as delivered, does not measurably change aggregate network outcomes. These findings suggest that in settings with persistent electricity debt, targeted financial relief may act more directly on the binding constraint than broad moral persuasion, while also underscoring the importance of measuring treatment reach and distinguishing recorded network losses from theft itself.

Online Appendix

A Feeder experimental sample and checks

Table A.1: Feeder-level pre-treatment summary statistics

	Mean	SD	Median	P25	P75	N
A) Feeder characteristics						
Total load (MW)	9.04	3.66	8.4	6.35	11	1926
Consumer Count (thousands)	4.69	1.85	4.43	3.35	5.58	1926
B) Energy flows and T&D losses						
Received Units (MWh)	1266	583	1150	829	1562	1926
Billed Units (MWh)	892	420	796	590	1084	1926
Adjusted Units (MWh)	-33.4	319	8.93	-27.1	33.7	1924
Net Billed Units (MWh)	858	525	791	558	1093	1926
T&D Loss (%)	.32	.274	.293	.236	.367	1926
C) Billing, payments and collection losses						
Assessed Amount (Million PKR)	23.9	16.6	20.5	13.7	31.3	1926
Payment (Million PKR)	20.5	12.1	17.6	12	26	1926
Collection Loss Billing Cycle (%)	.215	3.12	.163	.00161	.319	1926
Collection Loss Calendar Cycle (%)	-.0879	1.48	.0153	-.213	.19	672

Notes: Unless otherwise noted, the unit of analysis is feeder × month (administrative billing data). Table reports mean, SD, median, P25, P75, and N. Loss-rate variables are expressed as proportions (e.g., 0.32 denotes 32%), despite percent signs in original row labels.

Table A.2: Feeder-level baseline balance

	Control	Treated	Difference (T–C)	p-value	N
A) Feeder characteristics					
Total load (MW)	9.09	8.99	-.104	.924	42
Consumer Count (thousands)	4.69	4.68	-.011	.984	42
B) Energy flows and T&D losses					
Received Units (MWh)	1297	1233	-64	.558	42
Billed Units (MWh)	908	874	-33.5	.685	42
Adjusted Units (MWh)	-29.9	-36.9	-7.05	.58	42
Net Billed Units (MWh)	878	838	-40.6	.606	42
T&D Loss (%)	.319	.322	.00357	.784	42
C) Billing, payments and collection losses					
Assessed Amount (Million PKR)	24	23.8	-.175	.944	42
Payment (Million PKR)	20.5	20.6	.0629	.978	42
Collection Loss Billing Cycle (%)	.261	.169	-.0926	.525	42
Collection Loss Calendar Cycle (%)	-.171	-.0046	.167	.138	42

Notes: Unless otherwise noted, the unit of analysis is a feeder. Loss rates are shown as proportions (0.319 denotes 31.9%), not percentage-point units.

Table A.3: Sermon effects by implementation and follow-up period

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Received Units (MWh)	Billed Units (MWh)	T&D loss (%)	Assessed Amount (Million PKR)	Payment (Million PKR)	Collection loss B	Collection loss C (%)
Treated × Nov 2024–Jan 2025	57.7	24.2	.0224	-.0119	-.258	.0959	-.188
	(68.7)	(49.3)	(.042)	(1.11)	(1.14)	(.148)	(.116)
Treated × Feb 2025–Apr 2025	13.7	43	-.0526	.766	.658	.221	-.143
	(60.1)	(41.3)	(.0397)	(1.07)	(.912)	(.177)	(.117)
Treated × May 2025–Jul 2025	-55	20.4	-.0659	.191	-.423	.201	-.134
	(76.7)	(54.5)	(.0395)	(1.71)	(1.52)	(.166)	(.118)
Treated × Aug 2025–Sep 2025	-32.1	23.1	-.0207	.543	-.695	.129	-.149
	(93.1)	(63.5)	(.0371)	(2.04)	(1.86)	(.142)	(.124)
Observations	2388	2388	2388	2388	2388	2388	1134
R-squared	.803	.777	.419	.718	.821	.0444	.0986
Feeder FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered at feeder	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively.

Table A.4: Exploratory heterogeneity by mosque treatment intensity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Received Units (MWh)	Billed Units (MWh)	T&D loss (%)	Assessed Amount (Million PKR)	Payment (Million PKR)	Collection loss B	Collection loss C (%)
Heterogeneity by fraction of mosques treated (median split among treated feeders)							
Low-intensity effect: $1\{Treated\} \times 1\{Post\}$	20.6	69.3**	-.0477	1.36	.619	.222	-.152
	(80.6)	(31.6)	(.0491)	(1.23)	(1.18)	(.166)	(.111)
High-intensity effect: $1\{Treated\} \times 1\{Post\}$	-21.4	-9.49	-.0138	-.557	-.818	.113	-.156
	(75.5)	(63.5)	(.0314)	(1.38)	(1.29)	(.156)	(.121)
High–Low difference: $1\{Treated\} \times 1\{Post\} \times 1\{High\}$	-42	-78.8	.0338	-1.92	-1.44	-.109	-.00432
	(95.4)	(61.2)	(.0499)	(1.65)	(1.54)	(.126)	(.0572)
Observations	2388	2388	2388	2388	2388	2388	1134
R-squared	.803	.778	.418	.719	.821	.0444	.0986

Notes: Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively.

B Household sample, implementation and robustness

Table B.1: Household administrative summary statistics

	Mean	SD	Median	P25	P75	N
A) Customer characteristics						
Total load of the customer (KW)	1.18	0.39	1.00	1.00	1.00	25940
B) Consumption & Billing						
Electricity Units Consumed (KWh)	111.21	80.92	96.00	52.00	158.00	25940
Amount due within due date (PKR)	5988.35	13671.92	2174.00	1137.00	5732.00	25940
C) Payments & Collection						
Payment Made (PKR)	2925.39	4459.06	1458.00	129.00	3175.50	25940
Collection Loss (%)	0.25	0.44	0.00	0.00	0.63	25930
D) Arrears						
Arrears (PKR)	2752.57	11938.35	0.00	0.00	307.50	25940

Notes: Unless otherwise noted, the unit of analysis is customer id × month (administrative billing data). Table reports mean, SD, median, P25, P75, and N. Collection loss is expressed as a proportion.

Table B.2: Household administrative balance

	No-subsidy mean	Subsidy mean	Difference (T–C)	p-value	N
A) No-sermon households					
Total load of the customer (KW)	1.25	1.17	-0.08	0.023	602
Electricity Units Consumed (KWh)	109.81	113.33	3.53	0.323	602
Amount due within due date (PKR)	6503.67	5844.11	-659.57	0.381	602
Payment Made (PKR)	2933.22	3000.37	67.15	0.684	602
Collection Loss (%)	0.26	0.25	-0.01	0.628	602
Arrears (PKR)	3221.12	2526.26	-694.86	0.261	602
B) Sermon households					
Total load of the customer (KW)	1.20	1.20	-0.00	0.936	617
Electricity Units Consumed (KWh)	111.63	109.87	-1.76	0.604	617
Amount due within due date (PKR)	5616.77	6120.98	504.21	0.551	617
Payment Made (PKR)	2918.59	2817.24	-101.35	0.517	617
Collection Loss (%)	0.24	0.25	0.01	0.616	617
Arrears (PKR)	2406.39	2957.32	550.93	0.402	617

Notes: Unless otherwise noted, the unit of analysis is a household.

Table B.3: Household survey balance

	No-subsidy mean	Subsidy mean	Difference (T–C)	p-value	N
A) No-sermon households					
Uses any solar	0.12	0.12	-0.01	0.802	602
Hooked connection / kunda	0.04	0.02	-0.01	0.339	596
Religiosity index	-0.01	0.03	0.03	0.541	601
Share of HH attending Friday prayers	0.35	0.39	0.04	0.010	602
Monthly household income (PKR)	26600.74	24734.08	-1866.67	0.650	583
Monthly household expenses (PKR)	21774.48	21562.21	-212.26	0.837	578
Budget stress index	-0.03	0.01	0.04	0.579	569
House-quality index	-0.03	-0.15	-0.12	0.041	601
House is pakki / solid	0.91	0.86	-0.05	0.042	586
B) Sermon households					
Uses any solar	0.15	0.17	0.02	0.502	617
Hooked connection / kunda	0.03	0.01	-0.02	0.167	615
Religiosity index	-0.07	0.02	0.09	0.120	617
Share of HH attending Friday prayers	0.36	0.35	-0.01	0.530	617
Monthly household income (PKR)	38744.38	36249.35	-2495.03	0.651	615
Monthly household expenses (PKR)	34058.09	33072.40	-985.69	0.825	615
Budget stress index	-0.02	-0.01	0.01	0.784	615
House-quality index	0.06	0.07	0.00	0.942	617
House is pakki / solid	0.84	0.88	0.04	0.133	606

Notes: Unless otherwise noted, the unit of analysis is a household.

Table B.4: Verification of subsidy delivery and reported sermon exposure

	No mosque treatment		Mosque treatment		Sermon exposure
	Nov–Apr	May–Aug	Nov–Apr	May–Aug	Endline
Treatment-group mean	510	1844	459	1580	.456
Control-group mean	0	0	0	0	.406
Difference (treat − control)	510***	1844***	459***	1580***	.0496*
	(22.2)	(78)	(20.9)	(74.8)	(.0286)
Observations	602	602	617	631	1196
R-squared	.467	.482	.441	.428	.00251
D) Descriptive means among finance-control households only					
Mean current bill (PKR)	1771	4044	1667	3880	
Mean payable within due date (PKR)	11011	14208	8685	11285	

Notes: Each outcome is averaged within household over the post period and then compared across subsidy status. Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively.

Table B.5: Household post-period estimates with baseline covariates

	(1) Units (kWh)	(2) Payable (PKR)	(3) Payment (PKR)	(4) Coll. loss (%)	(5) Arrears (PKR)	(6) Solar adoption	(7) Hooked conn.
A) No mosque treatment: finance treated vs finance control							
Finance treated − finance control	4.75	-1615	-478***	-.0126	-675	.0111	-.00282
	(2.96)	(1761)	(159)	(.0182)	(1556)	(.0272)	(.0259)
Finance-control mean	105	12603	2857	.298	8351	.121	.1
Observations	561	561	561	561	561	553	532
R-squared	.387	.405	.374	.458	.367	.0712	.0134
Baseline survey controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-period mean of outcome	Yes	Yes	Yes	Yes	Yes	No	No
B) Mosque treatment: finance treated vs finance control							
Finance treated − finance control	1.65	-3826***	-520***	-.00933	-2713**	-.00725	-.00588
	(2.92)	(1340)	(158)	(.0162)	(1156)	(.0249)	(.0287)
Finance-control mean	104	9765	2791	.276	5890	.104	.136
Observations	613	613	613	613	613	599	577
R-squared	.323	.429	.326	.527	.399	.0249	.022
Baseline survey controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-period mean of outcome	Yes	Yes	Yes	Yes	Yes	No	No

Notes: Each outcome is averaged within household over the post period and then compared across subsidy status. Regressions additionally control for baseline solar use, baseline hooked connection/kunda, religiosity index, share of household attending Friday prayers, log monthly household income, log monthly household expenses, and the baseline house-quality index. Columns (1)-(5) also control for the household's own pre-period mean of the dependent variable. Columns (6)-(7) do not have a panel pre-period analogue; for those outcomes the baseline solar/kunda measures serve as the closest pre-treatment controls. Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively.

Table B.6: Subsidy effects by programme and follow-up period

	(1) Units (kWh)	(2) Payable (PKR)	(3) Payment (PKR)	(4) Coll. loss (%)	(5) Subsidy (PKR)	(6) Arrears (PKR)
A) No sermons: time-varying subsidy effect						
Subsidy × Nov 2024–Jan 2025	-.842	-3241*	-501***	-.0904*	541***	-2346
	(3.04)	(1704)	(190)	(.0469)	(25.2)	(1672)
Subsidy × Feb 2025–Apr 2025	-3.11	-3136*	-576***	-.0119	482***	-2435
	(3.54)	(1869)	(220)	(.0214)	(22.7)	(1833)
Subsidy × May 2025–Aug 2025	5.46	-2928	-1008***	.019	1845***	-1137
	(5.13)	(2102)	(266)	(.0208)	(78.1)	(1908)
Subsidy × Sep 2025–Nov 2025	6.55	-37	268	.0325	1.56	326
	(4.05)	(2260)	(232)	(.024)	(2.31)	(1917)
Observations	20616	20616	20616	20607	20616	20616
Within R-squared	.000634	.00217	.00204	.000987	.385	.00108
B) Sermons: time-varying subsidy effect						
Subsidy × Nov 2024–Jan 2025	2.17	-2921**	-401**	-.0398	482***	-2881**
	(3.09)	(1445)	(190)	(.0301)	(24)	(1330)
Subsidy × Feb 2025–Apr 2025	-1.35	-3337**	-247	-.000188	439***	-2220*
	(3.25)	(1348)	(244)	(.0214)	(21.2)	(1206)
Subsidy × May 2025–Aug 2025	2.25	-4960***	-1251***	-.00797	1569***	-2412*
	(4.77)	(1471)	(240)	(.0208)	(73.4)	(1253)
Subsidy × Sep 2025–Nov 2025	7.99*	-4693***	102	-.0323	1.85	-3851***
	(4.13)	(1650)	(253)	(.0212)	(1.86)	(1348)
Observations	21171	21171	21171	21168	21171	21171
Within R-squared	.000479	.00686	.00245	.000302	.334	.00348

Notes: Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively, and are shown only for the treatment effect.

C Additional household heterogeneity

These subgroup comparisons are exploratory, not confirmatory tests of mechanisms.

Table C.1: Exploratory heterogeneity by pre-treatment bills

Heterogeneity by persistent high-bill exposure

H=1 if current bill exceeds 6,667 PKR in at least 10% of pre-period months

	(1) Units (kWh)	(2) Payable (PKR)	(3) Payment (PKR)	(4) Coll. loss (%)	(5) Subsidy (PKR)	(6) Arrears (PKR)
A) No sermons						
H=0 effect: Subsidy × Post	1.06	-4382*	-394***	-.0247	547	-2966
	(3.36)	(2428)	(150)	(.0262)	(0)	(2274)
H=1 − H=0	5.43	4722	-157	.0321	578	3675
	(6.35)	(3560)	(339)	(.037)	(0)	(3286)
H=1 total effect	6.5	339	-552*	.0074	1125	709
	(5.39)	(2604)	(304)	(.0262)	(0)	(2372)
Pre-period mean, H=0	84.7	3905	1659	.257	0	2087
Pre-period mean, H=1	148	9029	4749	.261	0	3781
Observations	20616	20616	20616	20607	20616	20616
Within R-squared	.0535	.00806	.0316	.00216	.239	.00292
B) Sermons						
H=0 effect: Subsidy × Post	3.42	-467	-211*	-.0145	494	-67.5
	(3.21)	(884)	(123)	(.0198)	(0)	(627)
H=1 − H=0	-4.06	-8490***	-798**	-.00919	531	-6372**
	(6.89)	(3053)	(383)	(.0343)	(0)	(2612)
H=1 total effect	-.634	-8957***	-1009***	-.0237	1025	-6439**
	(6.09)	(2922)	(362)	(.028)	(0)	(2535)
Pre-period mean, H=0	86.2	3231	1644	.238	0	1470
Pre-period mean, H=1	148	9851	4718	.253	0	4518
Observations	21171	21171	21171	21168	21171	21171
Within R-squared	.0469	.0268	.0256	.00196	.203	.0158

Notes: Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively, and are shown only for the treatment effect.

Table C.2: Exploratory heterogeneity by budget stress

Heterogeneity by baseline budget stress

H=1 if log(expenses) − log(income) exceeds the baseline-sample median

	(1) Units (kWh)	(2) Payable (PKR)	(3) Payment (PKR)	(4) Coll. loss (%)	(5) Subsidy (PKR)	(6) Arrears (PKR)
A) No sermons						
H=0 effect: Subsidy × Post	1.81	-4981*	-530**	-.019	797	-3428
	(4.07)	(2722)	(211)	(.0274)	(0)	(2455)
H=1 − H=0	.796	5561	10.1	.0214	16.4	4436
	(6.73)	(3681)	(339)	(.0379)	(0)	(3474)
H=1 total effect	2.6	580	-520*	.00246	813	1008
	(5.36)	(2477)	(266)	(.0261)	(0)	(2458)
Pre-period mean, H=0	111	6580	2965	.271	0	3227
Pre-period mean, H=1	112	5535	2969	.251	0	2354
Observations	19461	19461	19461	19452	19461	19461
Within R-squared	.0015	.00591	.00188	.00139	.166	.00377
B) Sermons						
H=0 effect: Subsidy × Post	3.53	-5535***	-416**	-.0434**	712	-4162**
	(3.91)	(1871)	(208)	(.0197)	(0)	(1622)
H=1 − H=0	-3.23	5054**	-345	.0791**	-51.4	4508**
	(6.86)	(2336)	(346)	(.0342)	(0)	(1958)
H=1 total effect	.296	-481	-761***	.0356	660	346
	(5.64)	(1399)	(277)	(.028)	(0)	(1097)
Pre-period mean, H=0	111	5850	2821	.238	0	2663
Pre-period mean, H=1	112	5916	3018	.255	0	2677
Observations	21104	21104	21104	21101	21104	21104
Within R-squared	.0021	.0115	.00287	.00235	.151	.00748

Notes: Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively, and are shown only for the treatment effect.

Table C.3: Exploratory heterogeneity by Friday-prayer attendance

Heterogeneity by Friday-prayer exposure

H=1 if share of household attending Friday prayers exceeds the baseline-sample median

	(1) Units (kWh)	(2) Payable (PKR)	(3) Payment (PKR)	(4) Coll. loss (%)	(5) Subsidy (PKR)	(6) Arrears (PKR)
A) No sermons						
H=0 effect: Subsidy × Post	5.2	-2335	-382*	.0224	807	-1984
	(4.8)	(2573)	(225)	(.0231)	(0)	(2355)
H=1 − H=0	-5.15	-140	-235	-.0626*	-5.45	1169
	(6.33)	(3611)	(323)	(.0362)	(0)	(3320)
H=1 total effect	.0501	-2475	-617***	-.0402	802	-815
	(4.13)	(2534)	(232)	(.0279)	(0)	(2340)
Pre-period mean, H=0	110	5620	2950	.264	0	2433
Pre-period mean, H=1	113	6512	2976	.254	0	3169
Observations	20616	20616	20616	20607	20616	20616
Within R-squared	.0013	.00242	.00344	.00199	.168	.00182
B) Sermons						
H=0 effect: Subsidy × Post	-1.21	-4319**	-437**	-.0269	683	-3051*
	(4.54)	(1837)	(203)	(.0211)	(0)	(1629)
H=1 − H=0	9.33	588	-160	.0183	30.1	524
	(6.23)	(2720)	(340)	(.0332)	(0)	(2269)
H=1 total effect	8.11*	-3732*	-597**	-.00859	713	-2527
	(4.27)	(2005)	(273)	(.0256)	(0)	(1579)
Pre-period mean, H=0	113	6123	2943	.248	0	2823
Pre-period mean, H=1	109	5627	2816	.239	0	2545
Observations	21171	21171	21171	21168	21171	21171
Within R-squared	.00225	.00773	.00233	.00139	.149	.00492

Notes: Standard errors are clustered at the customer level. Robust standard errors in parentheses. The 90%, 95%, and 99% significance levels correspond with *, **, and *** respectively, and are shown only for the treatment effect.

D Selected event-study evidence

The figures report event-study estimates for the feeder and household experiments.

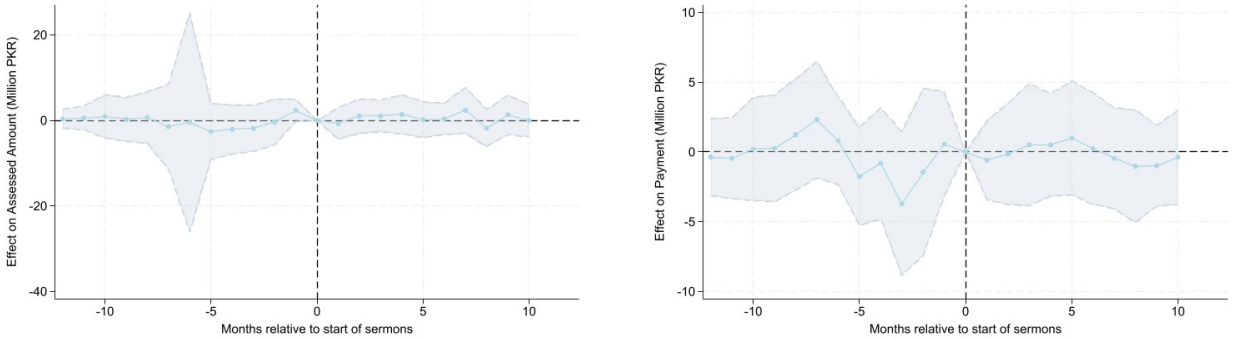


Figure D.1: Sermon assignment: billed assessments and payments

Notes: Feeder event-study coefficients for assessed amounts (left) and payments (right), with the intervention start marked by the dashed vertical line. Coefficients are normalised to the intervention month.

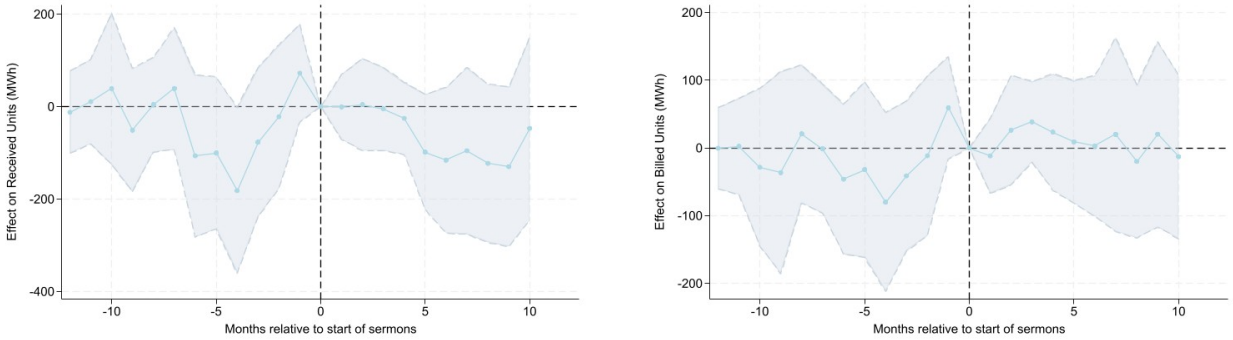


Figure D.2: Sermon assignment: electricity received and billed

Notes: Feeder event studies of energy received (left) and billed (right). These administrative outcomes do not separately identify unauthorised consumption.

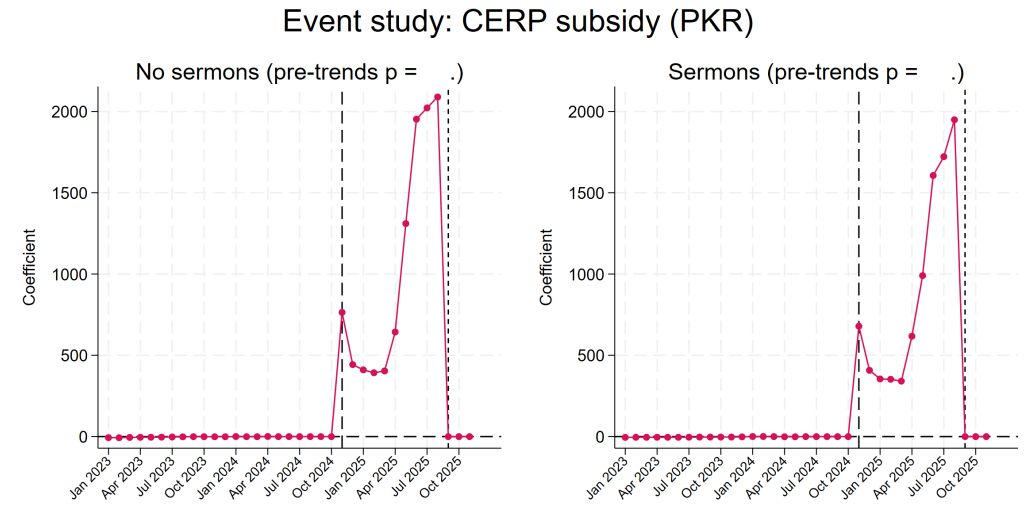


Figure D.3: Household subsidy assignment: subsidy credit received

Notes: Household event study, estimated separately by sermon environment. October 2024 is the reference month. The plot depicts administrative subsidy credits, not consumption.

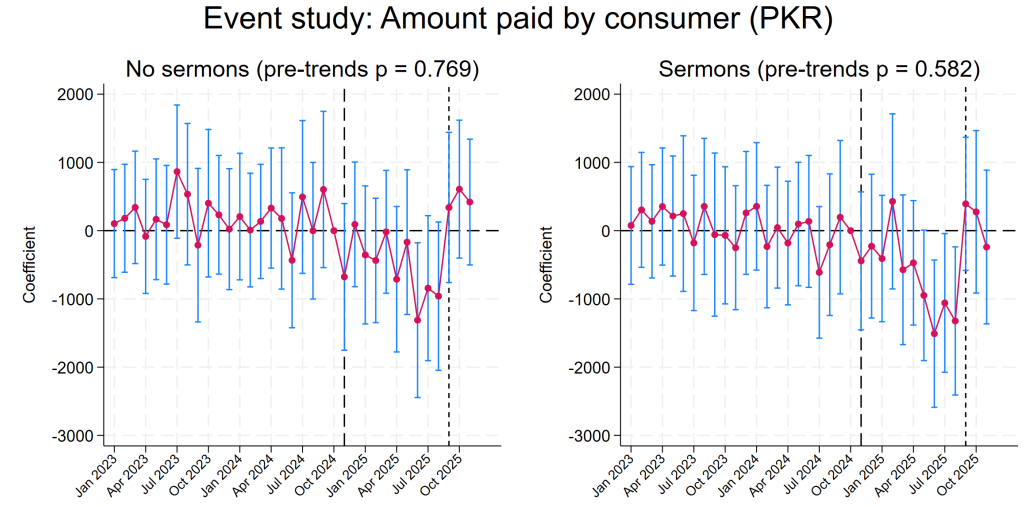


Figure D.4: Household subsidy assignment: amount paid by the consumer

Notes: Household event study, estimated separately by sermon environment, relative to October 2024. Error bars denote 95 percent confidence intervals.

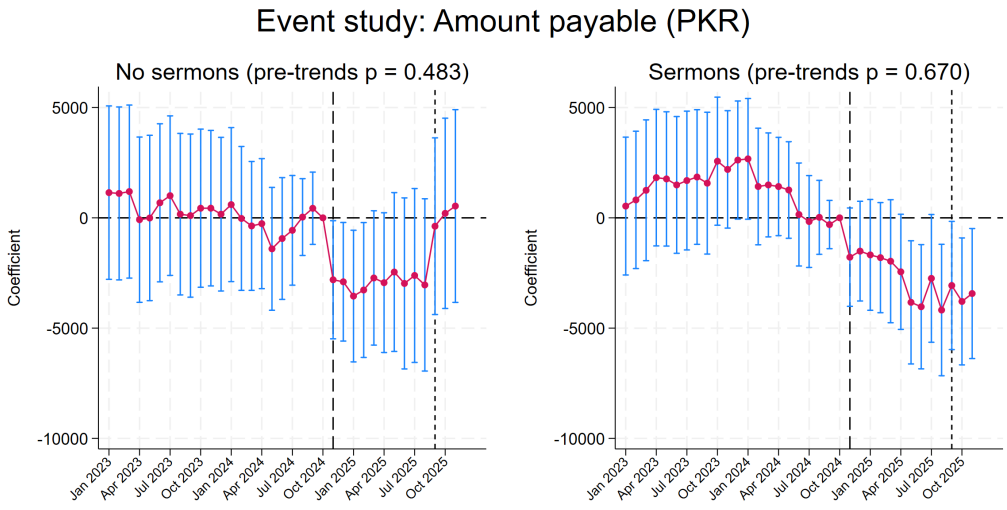


Figure D.5: Household subsidy assignment: amount payable

Notes: Household event study, estimated separately by sermon environment, relative to October 2024. Error bars denote 95 percent confidence intervals.

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